

IDENTIFYING SUSPICIOUS PROCESSES USING SYSINTERNALS PROCESS EXPLORER

EX. NO: 9

Date :

1. AIM

To examine active Windows processes using Sysinternals Process Explorer, analyze process hierarchy, executable properties, digital signatures, execution paths, resource usage, thread information, and threat-intelligence results to identify potentially suspicious processes.

2. SOFTWARE / TOOLS REQUIRED

Item	Details
Operating System	Windows
Forensic Tool	Sysinternals Process Explorer
Executable	procexp64.exe
Execution Privilege	Administrator
Process Analysis	Process tree and process properties
Threat Intelligence	VirusTotal
Security Tool	Windows Security / Microsoft Defender

Utilities / Components Used

- **Process Explorer** – Used to view and analyze active Windows processes.
- **Process Properties** – Used to examine process paths, descriptions, signatures, and execution information.
- **Thread Information** – Used to examine threads associated with a process.
- **VirusTotal** – Used to verify the reputation of a selected executable.
- **Windows Security** – Used to verify the system's threat-protection status.

3. THEORY

Sysinternals Process Explorer is an advanced Windows process-monitoring utility that provides detailed information about running processes.

It displays processes in a hierarchical parent-child structure and provides information such as Process ID, CPU usage, memory usage, description, company name, executable path, and process properties.

Step 1: Launch Process Explorer

The main interface was examined to view the active processes and available process information.



The process tree was examined to understand the parent-child relationships between running processes.

The hierarchy provides a structured view of how processes are created and related to one another.



Step 3: Examine Process Properties

A selected process was opened in the Properties window.

The Image section was examined for executable information such as the description, path, command line, user information, and process details.

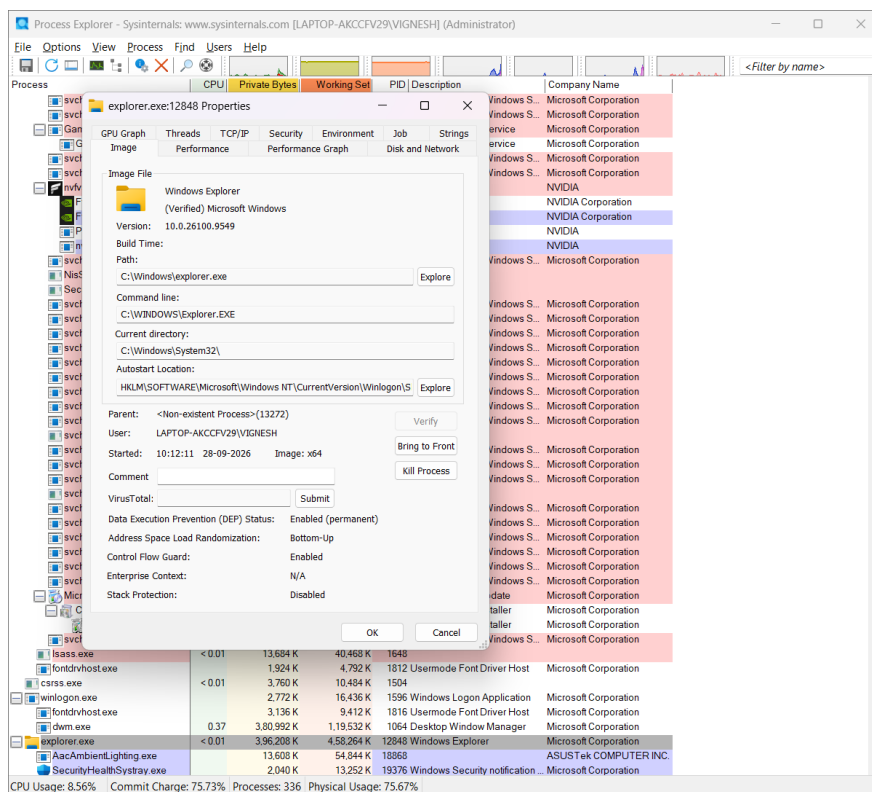


Figure 3: Process Properties window displaying executable and process information.

Step 4: Verify Digital Signature

The selected process was examined for publisher and digital-signature information.

Signature verification helps determine whether an executable is associated with a trusted software publisher.

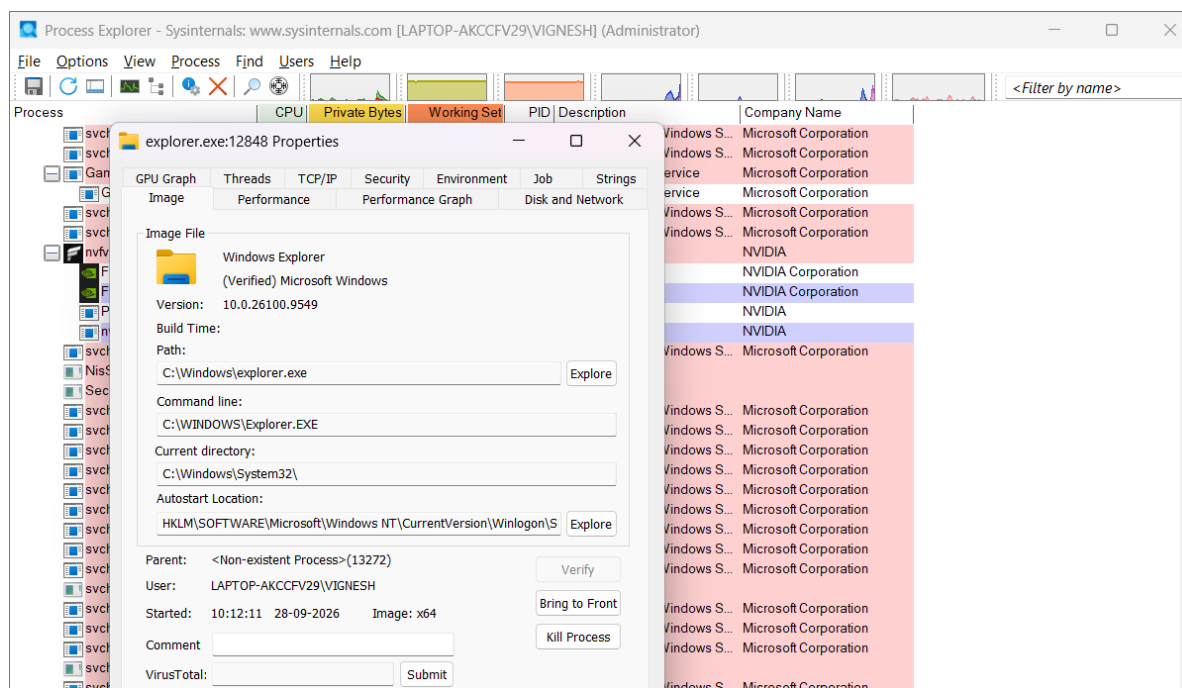


Figure 4: Verification of the digital signature and publisher information of the selected process.

Step 5: Verify Execution Path

The executable path of the selected process was examined. The location of an executable can provide useful information when investigating potentially suspicious processes. Unexpected locations may require additional examination.

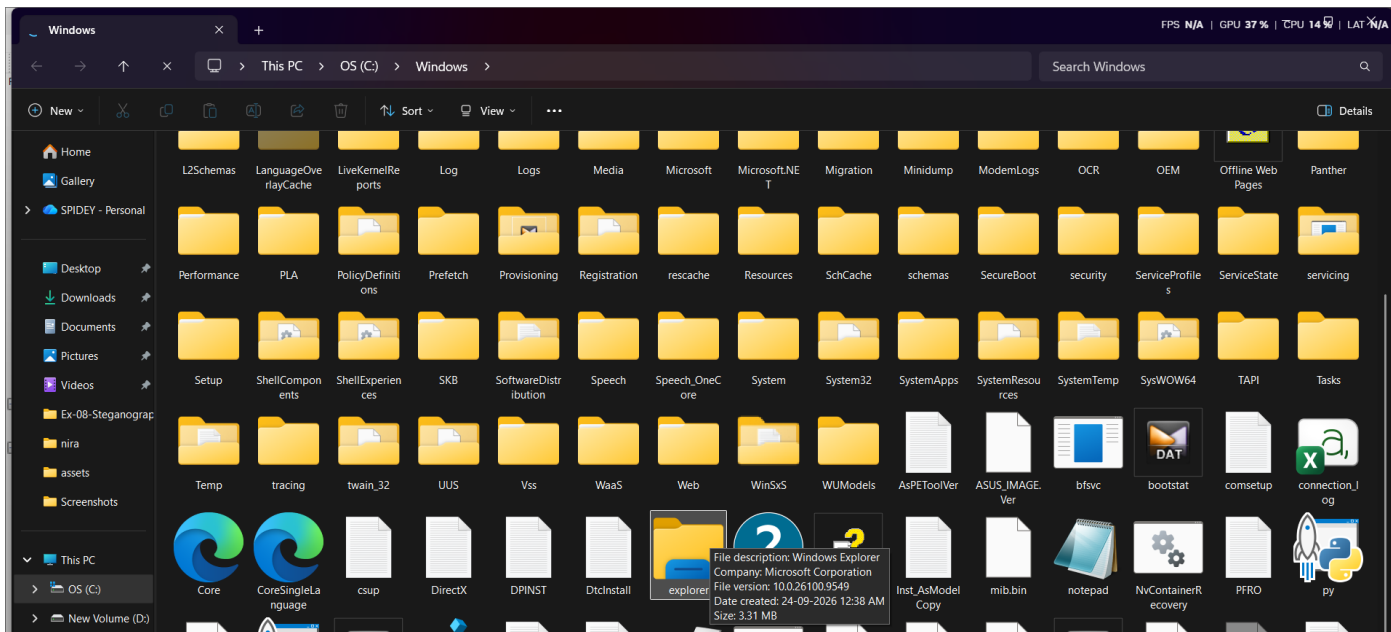


Figure 5: Verification of the executable path of the selected process.

Step 6: Examine Thread Information

The Threads section of the Process Properties window was examined.

The available information includes thread identifiers, CPU activity, start addresses, execution state, and other thread-related details.

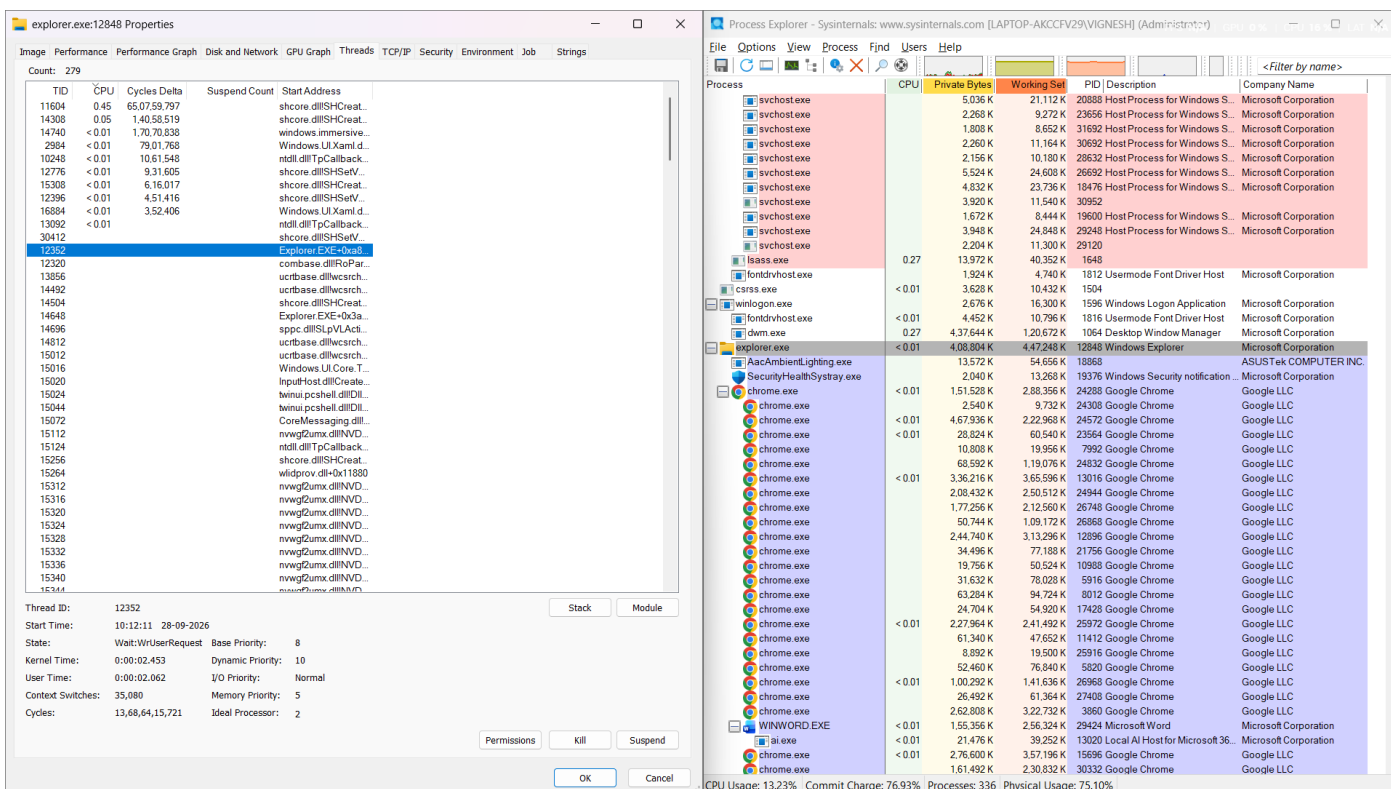


Figure 6: Thread information associated with the selected process.

Step 7: Perform Threat-Intelligence Verification

The selected executable was examined using VirusTotal to obtain additional security information.

The result was reviewed to determine whether security vendors had identified the analyzed executable as malicious.

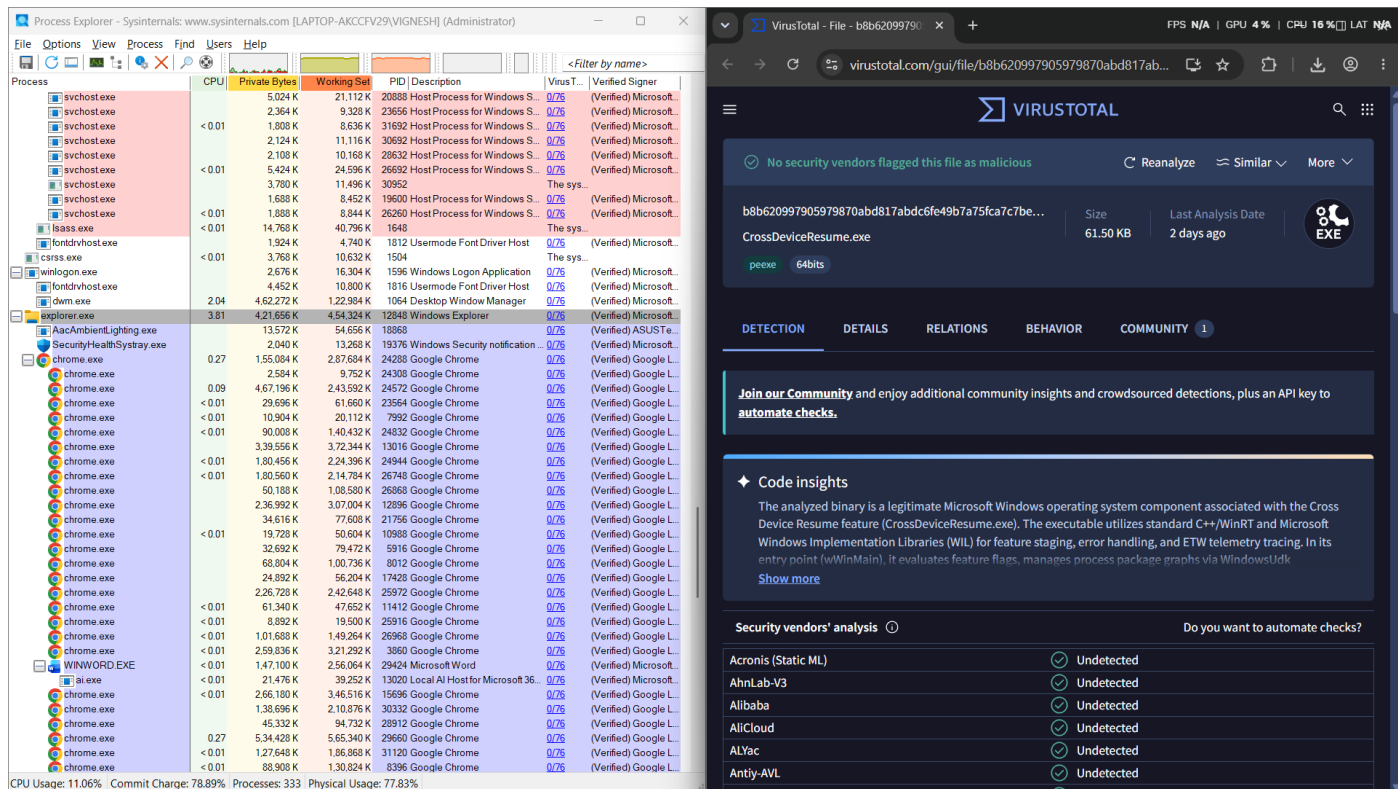


Figure 7: VirusTotal threat-intelligence analysis of the selected executable.

Step 8: Verify System Security Status

Windows Security was opened to examine the current virus and threat-protection status.

The available scan and protection information was reviewed to determine the current security status of the system.

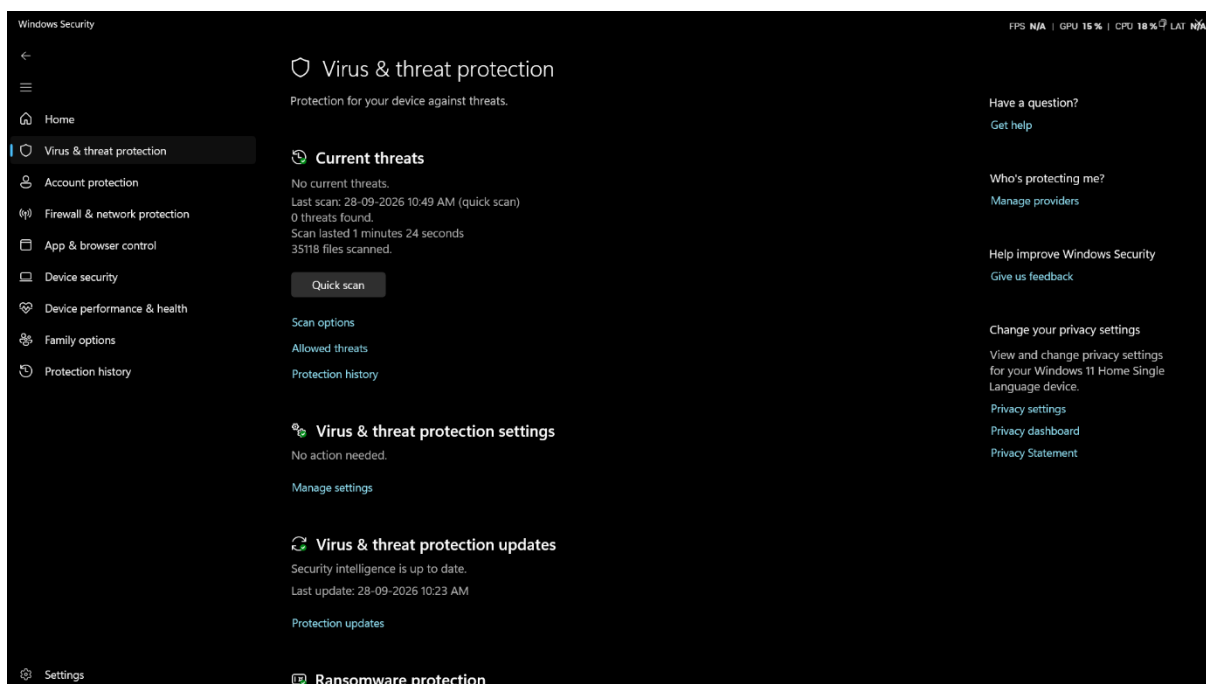


Figure 8: Windows Security displaying the system's virus and threat-protection status.

Step 9: Security Response

If a process is confirmed to be malicious after proper investigation, Process Explorer can be used to suspend or terminate the process.

The source executable should only be removed after appropriate verification and authorization.

A security scan should then be performed using Windows Defender or another trusted anti-malware solution.

5. OBSERVATIONS / FORENSIC FINDINGS

1. Process Explorer provided a detailed view of active Windows processes.
2. The process hierarchy allowed parent-child relationships to be examined.
3. Process Properties provided detailed executable and security information.
4. Digital-signature information could be examined for process verification.
5. Executable paths could be inspected for unusual locations.
6. Thread information provided additional details about process execution.
7. Threat-intelligence verification provided additional information about executable reputation.
8. Windows Security provided information about the current system threat-protection status.
9. Process Explorer can support preliminary identification and investigation of potentially suspicious processes.

6. LIMITATION

Process Explorer provides process-level information but cannot by itself conclusively determine whether a process is malicious. Additional forensic analysis, file verification, threat-intelligence sources, and security scanning may be required before taking containment or removal actions.

7. CONCLUSION

The experiment demonstrated the use of Sysinternals Process Explorer for monitoring and examining active Windows processes. Process hierarchy, process properties, digital signatures, execution paths, thread information, threat-intelligence results, and system security status were examined as part of the process-investigation workflow. The experiment provided practical knowledge of using Process Explorer for process monitoring and preliminary identification of potentially suspicious activity.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

8.RESULT

The experiment successfully demonstrated the use of Sysinternals Process Explorer for examining active processes, process hierarchy, executable properties, digital signatures, execution paths, thread information, and security indicators. The obtained observations were documented through the corresponding figures.